



Session 24

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Module 4, Week 11
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Module 4: **Core Supervised Learning**



Regression and Classification

Using

Sklearn (A tool kit/ML-package in python)

<https://colab.research.google.com/drive/1DxRG2Q9mSYhRilZS7qhP8GGlwVe6j6Wg?usp=sharing>

Objectives



- Understand the concept of Multiple Linear Regression (MLR).
- Differentiate between Simple and Multiple Linear Regression.
- Build a Multiple Linear Regression model using Scikit-learn.
- Train and evaluate a regression model with multiple predictors.
- Interpret regression coefficients and model performance.

Why Multiple Linear Regression?

- Real world is little more complex
- One output variable can depend on many input variables.
- For eg.,
 - House price** can depend on *number of bed rooms, age of the house, size of the house, number of amenities around the society, location of the area, etc.*
 - Student Performance** in coming exams may depend on *Study Hours, Attendance in the course, Previous GPA, etc.*
 - Salary Prediction** for a new joinee may depend on *Experience, Education, Leadership score, other Skills.*

Multiple Linear Regression allows us to model multiple predictors/input variables

Simple Linear Regression (SLR) Vs Multiple Linear Regression (MLR)



- If multiple predictors are there, Multiple Linear Regression may **assure more accuracy** in prediction.
- Simple linear regression **can be visualized**, while MLR is more than two dimensions.
- Simple linear regression is the **foundation to understand** the concept of regression algorithm, MLR is built on top of the idea of simple linear regression.
- More features/predictors assures more accurate system, hence MLR is preferable over SLR if we have more features.

Modeling multiple linear regression


$$y = mx + c \text{ or } y = f(x)$$

// y is a function of x with unknown parameters m and c , where m is the slope (coefficient) of the line, c is the intercept (value of y at $x=0$)

$$y = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + \dots w_nx_n \text{ or } y = f(x_0, x_1, x_2, \dots, x_n) // m$$

and c are represented as weight (w) parameters in ML

where,

- y = dependent variable
- $x_1, x_2, \dots, x_n$ = independent variables/predictors
- w_0 = intercept
- $w_1, w_2, w_3, \dots, w_n$ = coefficients/weights

Modeling multiple linear regression


$$\text{Salary} = 25000 + 4000 * (\text{Experience}) + 3000 * (\text{Education}) + 500 * (\text{CS})$$

$$w0 = 25000 \text{ (base salary without any } x), w1 = 4000, w2 = 3000, w3 = 500$$

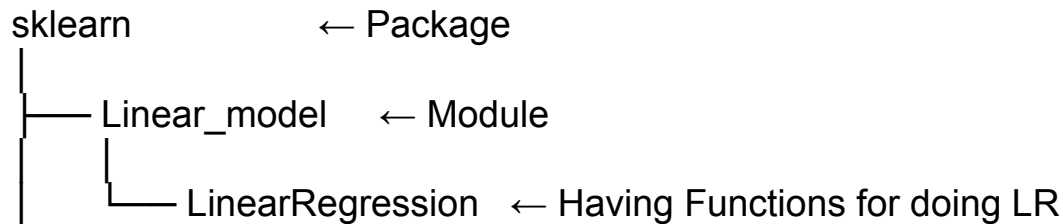
If input is:

Experience (x_1) = 5 years, Education (x_2) = 2 (Master's), Communication skill score out of 100 (x_3) = 30

$$\text{Predicted Salary} = 25000 + 4000(5) + 3000(2) + 500(30) = 25000 + 20000 + 6000 + 15000 = 66000$$

- Each feature contributes independently to the final prediction in proportion of the weights.
- The job of multiple linear regression algorithm is to learn optimal values of weights/coefficients

How to implement LR in Python?



```
from sklearn.linear model import LinearRegression
```

- Sklearn Package (library) is The Scikit-learn machine learning package in python.
- Linear_model is a module inside sklearn containing linear machine learning models.
- LinearRegression is a class for creating Linear Regression model objects



Practice Example //Optional slide

$[w_1 = 7509.57061562, w_2 = -785.30781169, w_3 = 104.500776]$

$w_0 = 18600.103466114895$

Predicted_salary (9th data point) = $w_0 + w_1 * 9 + w_2 * 3 + w_3 * 48 = 88846$

Test data point = 9th point, with experience of 9 years and salary is 90000

Errors and their interpretation



Lower MAE $\rightarrow$ Better predictions

Lower MSE $\rightarrow$ Smaller prediction errors

R^2 closer to 1 $\rightarrow$ Better fit

Performance Evaluation



Three suitable metrics

1. Mean Absolute Error (MAE): Measures average absolute prediction error, lower is better.
2. *Mean Square Error (MSE): Measure average square prediction error, lower is better // suitable for finding optimal parameter values also

Root Mean Squared Error (RMSE): Take root of the 2

3. R^2 Score: Measures goodness of fit between actual y and predicted y , higher is better, maximum value is 1.

Limitations of LR



- Will fail for data having complex relationship with y , other more sophisticated algorithms are the solution
- Outliers may spoil the model



Thank you!